Introduction to data tables and data frames using Python (Pandas)

Learning Objectives

By the end of this session, you will be able to:

- Understand **Data Tables and DataFrames** in Python.
- Learn how to create, manipulate, and analyze data using **pandas** and **NumPy**.
- Explore key operations like indexing, filtering, aggregation, and merging.

Different Data Types: Quick Reference

Data Types	Definition
List	An **ordered, mutable collection** of elements that allows duplicates. Defined using [ ].
Tuple	An **ordered, immutable collection** of elements. Defined using ( ).
Dictionary	A **key-value pair collection** that is unordered and mutable. Defined using { }.
Array (NumPy)	A **fixed-type collection of elements** optimized for numerical computations. Requires NumPy.
Series (Pandas)	A **one-dimensional labeled array** where each element has an index. Used in Pandas for data analysis.
DataFrame (Pandas)	A **two-dimensional table-like data structure** with labeled rows and columns. Used for structured data manipulation.

What is a Data Table?

- A Data Table is a structured way to store and organize data in a tabular format with rows and columns.
- It allows for easy access, manipulation, and analysis of data.
- Commonly used in databases (SQL), spreadsheets (Excel), and programming languages (Python Pandas, R, etc.).
- **Used in Python through Pandas DataFrames.**

Example:

ID	Name	Age	Salary
1	Alice	25	50,000
2	Bob	30	60,000
3	Charlie	35	70,000

What is a DataFrame?

- A two-dimensional labeled data structure in Pandas.
- Columns can have different data types (e.g., int, float, string).
- It's similar to a SQL table or a spreadsheet in Excel.

Mathematical Representation:

A DataFrame D with m rows and n columns can be represented as:

$$D = \begin{bmatrix} d_{11} & d_{12} & \dots & d_{1n} \\ d_{21} & d_{22} & \dots & d_{2n} \\ \dots & \dots & \dots & \dots \\ d_{m1} & d_{m2} & \dots & d_{mn} \end{bmatrix}$$

DataFrames in Python

- Creating DataFrames in Python
- Data Operations on DataFrames
- Modifying Data in DataFrames
- Aggregations and Summary Statistics
- Merging and Joining DataFrames
- Handling Missing Values.

Multiple Choice Questions (MCQs)

Q1: What is the primary difference between a Python list and a Pandas DataFrame?

A. Lists store only numbers, DataFrames store only text.

B. Lists are 1D, DataFrames are 2D.

C. Lists are used only in machine learning.

D. DataFrames cannot store numerical data.

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Fill in the Blanks

1. A DataFrame is a _____ data structure used for data manipulation in Python.
2. In a Data Table, the horizontal sections are called _____, and the vertical sections are called _____.
3. A dictionary in Python stores data as _____ pairs.
4. A tuple is similar to a list, but it is _____, meaning its values cannot be changed.
5. The process of summarizing and organizing data to extract useful insights is called _____.
6. The vertical sections in a DataFrame, each representing a specific attribute or feature, are called _____.
7. A structured collection of data that is stored electronically and can be queried is known as a _____.
8. The function used to remove missing values from a DataFrame is _____.

Fill in the Blanks

1. A DataFrame is a **two-dimensional** data structure used for data manipulation in Python.
2. In a Data Table, the horizontal sections are called **rows**, and the vertical sections are called **columns**.
3. A dictionary in Python stores data as **key-value** pairs.
4. A tuple is similar to a list, but it is **immutable**, meaning its values cannot be changed.
5. The process of summarizing and organizing data to extract useful insights is called **data analysis**.
6. The vertical sections in a DataFrame, each representing a specific attribute or feature, are called **columns**.
7. A structured collection of data that is stored electronically and can be queried is known as a **database**.
8. The function used to remove missing values from a DataFrame is **dropna()**.

Session Summary

In this session, we have discussed about:

- DataFrames are essential for structured data manipulation.
- Use Pandas for efficient handling of datasets.
- Practice coding problems to master DataFrame operations.